

16 • Understand & Measure the Decode Phase

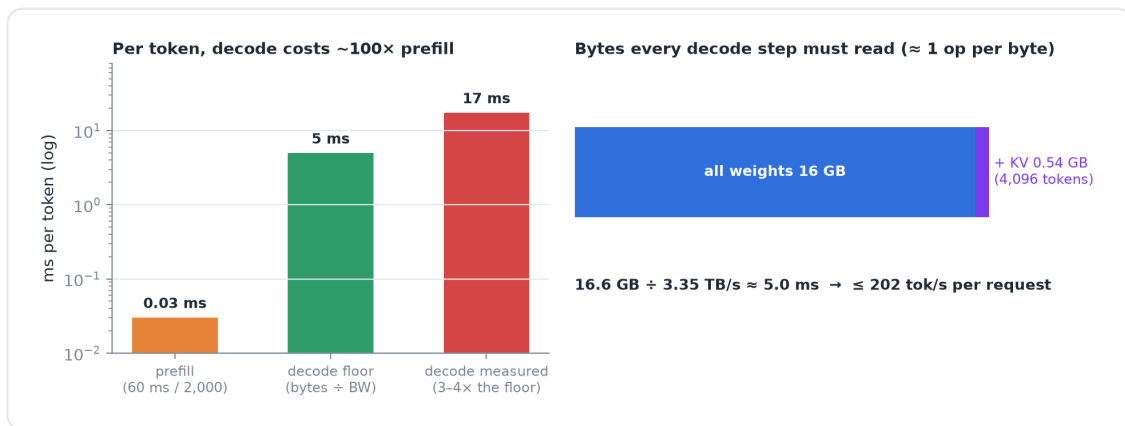
Source: [fanout.sh / inference-eng / 03-03](https://fanout.sh/inference-eng/03-03) · ~6 min

IN SHORT

Prefill handles 2,000 tokens in ~60 ms; decode then spends **several ms on every single token**, so per token it can be **~100× more expensive**. Each decode step pushes the newest token through all 32 layers **alone** (vector × matrix) and must read **all the weights plus this request's whole KV cache**, doing ~1 op per byte: **memory-bound**. So **time per token ≥ bytes ÷ bandwidth**: $\sim 16.6 \text{ GB} \div 3.35 \text{ TB/s} \approx 5 \text{ ms}$ ($\leq \sim 200$ tokens/s per request), and real batch-1 systems land **3–4× above** that floor. **Batching** shares the weight read: batch 64 is 15 ms per step but **>4,000 tokens/s**, until the **KV cache fills memory**. Measure with a short prompt, a forced long output, TPOT, the inter-token gap tail, and a concurrency sweep.

1. One decode step

The newest token goes through all 32 layers **by itself**. In each layer it meets every weight matrix, but as **one vector times a matrix**, not a block of tokens. In attention, its query reads the **keys and values of every earlier token** from the cache. One token comes out, its K and V join the cache, and the next step begins.



Every step reads two things: all the weights and this request's whole KV cache, with ~1 op per byte against the H100's ~300 ridge.

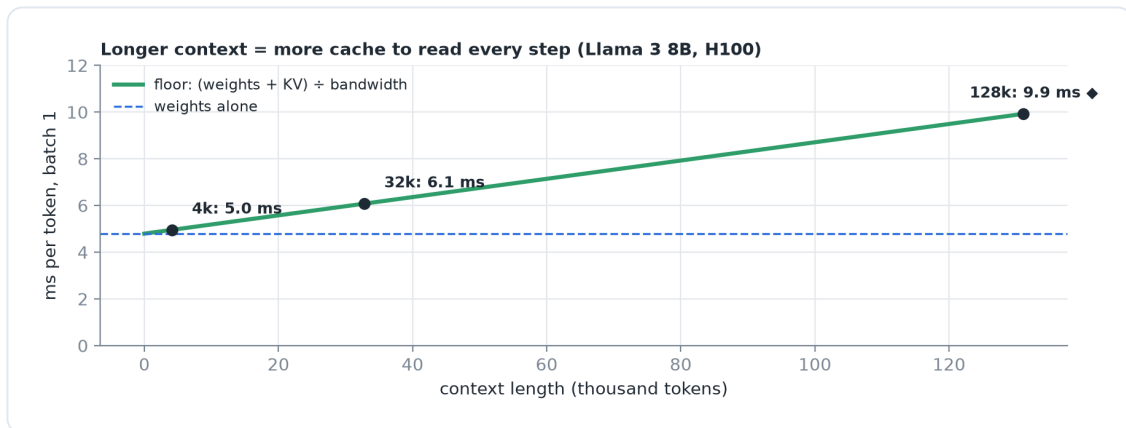
KEY IDEA

Decode is **memory-bound**: the GPU spends its time waiting for bytes, so its speed limit is **bytes read ÷ memory bandwidth**.

2. The speed limit

- Llama 3 8B weights ≈ **16 GB**; a 4,096-token conversation adds ≈ **0.5 GB** of KV cache.

- $16.6 \text{ GB} \div 3.35 \text{ TB/s} \approx \mathbf{5 \text{ ms per token}}$: one request can't beat $\sim \mathbf{200 \text{ tokens/s}}$ on this GPU.
- Longer context means more cache to read: at $\mathbf{32k \text{ tokens}}$ the floor rises to $\sim \mathbf{6 \text{ ms}}$.

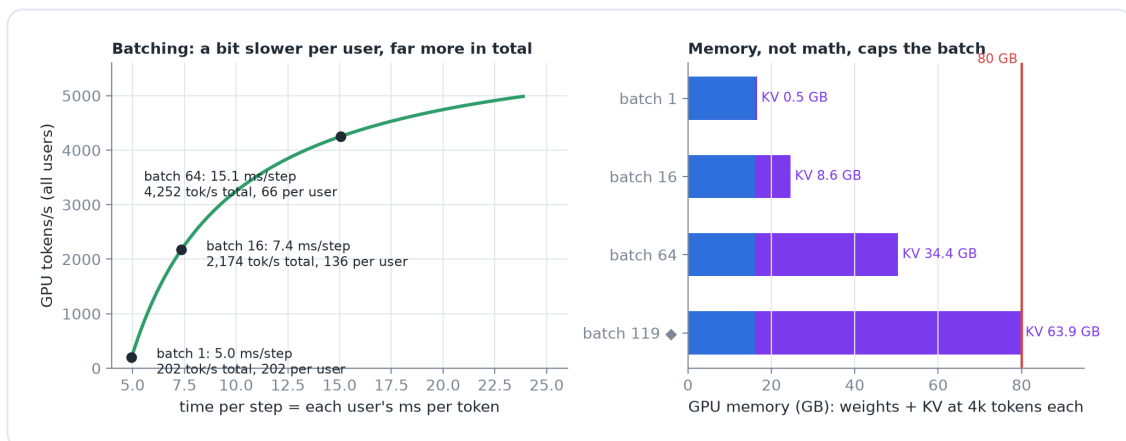


TRAP

Real systems land **above** the floor. One recent study measured batch-1 decode of 7–8B models on H100s at **a quarter to a third** of this ideal: small kernels and launch overheads eat the time. The floor tells you **where the ceiling is, not where you are**.

3. Batching: how servers reach thousands of tokens/s

If 16 requests decode together, the **weights are read once for all 16**. Each request still reads its own KV cache, but the big 16 GB read is **shared**.



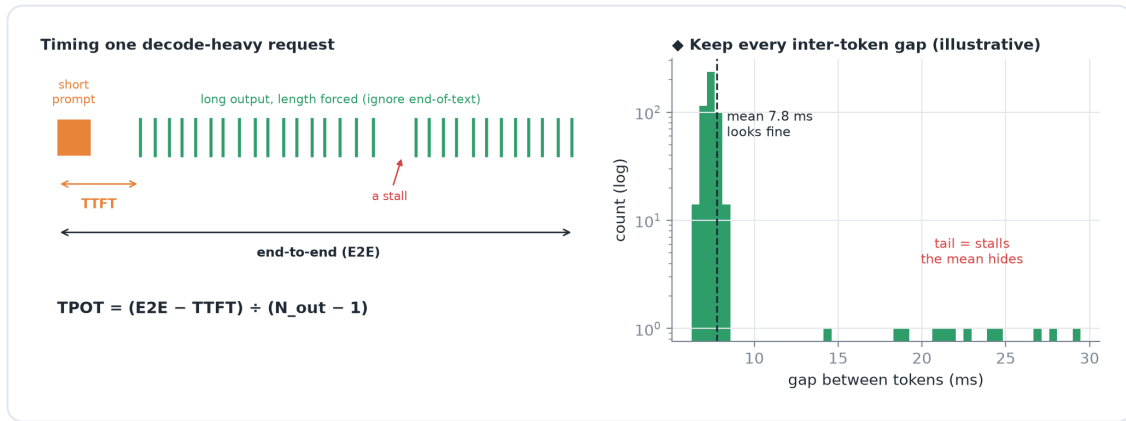
Left: each user's stream slows a little; the GPU does far more. Right: every request keeps its KV cache in memory; 64 requests $\times$ 4k tokens $\approx$ 34 GB on top of 16 GB of weights. ♦ At 4k each, $\sim$ 119 fit in 80 GB (see 02).

INTUITION

Decode's central trade-off is **latency per user vs total throughput**, and **memory, not math, caps the batch**.

4. Measure it honestly

- **Short prompt, long output** (a few hundred tokens), so prefill barely matters.
- **Force the length** (e.g. ignore the end-of-text token) so every run generates the same count.
- Record **TTFT** and **end-to-end** time, then $\text{TPOT} = (\text{E2E} - \text{TTFT}) \div (\text{N}_{\text{out}} - 1)$.
- Keep the **individual inter-token gaps**: their tail shows stalls an average hides.



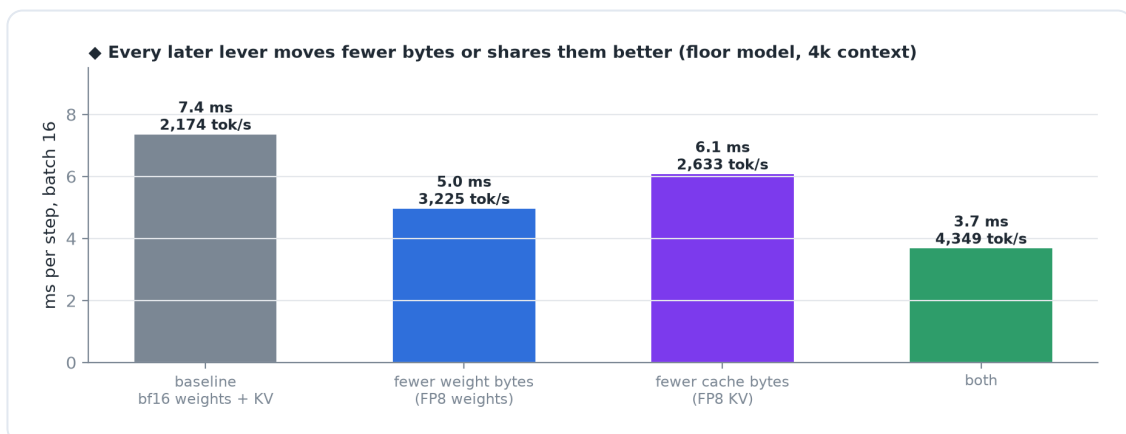
Then **sweep concurrency** (1, 2, 4, 8 ... 64), plot throughput against time per token, and convert to **achieved bandwidth** = bytes per step ÷ measured step time, e.g. $16.6 \text{ GB} \div 12 \text{ ms} \approx 1.4 \text{ TB/s} \approx 41\%$ of peak.

CAUTION

GPU utilization can read **~100% during decode while the math units are mostly idle**. It only means *some kernel was running* (see 09).

5. Every later lever

Quantization, cache compression and smarter batching all **move fewer bytes or share them better**.



◆ Floor-model arithmetic at batch 16, 4k context: halving weight bytes helps most here; halving cache bytes matters more as batch or context grows.

6. Recap

Lesson 16 on one card

ONE STEP

one token per request, alone
rereads all weights + its KV cache
 ≈ 1 op/byte: memory-bound

SPEED LIMIT

time $\geq$ bytes $\div$ bandwidth
 $16.6 \text{ GB} \div 3.35 \text{ TB/s} \approx 5 \text{ ms}$ ($\leq 200 \text{ tok/s}$)
real batch 1: 3-4 $\times$ the floor

BATCHING

weights read once for the batch
16: 7.4 ms, 2.2k tok/s; 64: 15 ms, 4.3k
until KV fills memory

MEASURE

short prompt, forced long output
 $\text{TPOT} = (\text{E2E} - \text{TTFT}) / (N - 1)$, gap tail
achieved BW; util $\neq$ busy math

Interview cheat sheet

Revise from these pages alone. ♦ = my addition beyond the video; everything else is from the lesson. The weights-only floor is in 01 (derivations 1 and 4 for the batched step), KV capacity in 02, launch overhead in 02, TTFT/TPOT request time in 08, MBU and utilization in 09.

Say it in 30 seconds

Decode runs one token per request through every layer as vector-times-matrix, rereading all weights plus that request's KV cache at about one op per byte, so it's memory-bound and time per token is at least bytes over bandwidth: about 5 ms for Llama 3 8B at 4k context on an H100, or 200 tokens/s, and real batch-1 runs are 3–4× slower. Batching shares the weight read, so batch 64 gives 15 ms per step but over 4,000 tokens/s, until KV memory runs out. Measure with a short prompt and forced long output, compute TPOT and the inter-token gap tail, sweep concurrency, and report achieved bandwidth, not GPU utilization.

Only numbers worth memorizing

- **Llama 3 8B, H100, 4k context:** floor $\approx$ **5 ms/token** ($\leq$ 200 tok/s); batch 16 $\approx$ **7.4 ms, 2.2k tok/s**; batch 64 $\approx$ **15 ms, 4.3k tok/s**, 34 GB of KV.

Derive, don't memorize

1. Decode vs prefill, per token

$$\frac{60 \text{ ms}}{2000 \text{ tokens}} = 0.03 \text{ ms} \quad \text{vs} \quad \approx 5 \text{ ms} \quad \Rightarrow \quad 100 \times \text{ or more}$$

2. The floor includes this request's cache

$$t \geq \frac{W + C \cdot \text{KV}}{\text{BW}} = \frac{16.06 \text{ GB} + 4096 \times 128 \text{ KB}}{3.35 \text{ TB/s}} = \frac{16.6 \text{ GB}}{3.35 \text{ TB/s}} \approx 5.0 \text{ ms} \quad C = 32\text{k} : \approx 6.1 \text{ ms}$$

INTUITION

The weights dominate until the cache rivals them: ♦ at batch 1 that takes $\sim$ 122k tokens (05). Real kernels add small-kernel and launch costs, landing 3–4× above.

3. Batching: throughput = batch ÷ step time

$$\frac{B}{t(B)} : \quad \frac{16}{7.4 \text{ ms}} \approx 2170 \text{ tok/s} \quad \frac{64}{15.1 \text{ ms}} \approx 4250 \text{ tok/s} \quad \text{KV} = 64 \times 0.54 \text{ GB} \approx 34 \text{ GB}$$

INTUITION

Per user, tokens/s = $1 \div t(B)$: $200 \rightarrow 136 \rightarrow 66$. Total goes up 21× while each user slows $\sim$ 3×.

Batch size vs speed (floor model, 4k context)

Batch	Step time	Per user	GPU total	KV in memory
1	5.0 ms	202 tok/s	202 tok/s	0.5 GB
16	7.4 ms	136 tok/s	2,174 tok/s	8.6 GB
64	15.1 ms	66 tok/s	4,252 tok/s	34 GB

Measurement recipe

Do	Why
Short prompt, long output	Prefill barely matters
Force the output length (ignore EOS)	Every run generates the same count
$TPOT = (E2E - TTFT) \div (N_{out} - 1)$	Decode pace without the first-token pause
Keep every inter-token gap	The tail shows stalls the mean hides
Sweep concurrency 1 → 64	Traces the latency vs throughput curve
Report achieved bandwidth	Bytes per step ÷ step time: distance to the floor

Symptom → diagnosis

Symptom	Diagnosis
GPU util ~100% but decode is slow	Util only means a kernel ran; check achieved bandwidth
Batch-1 TPOT 3-4× the floor	Small kernels and launch overheads
Mean TPOT fine, users see stutters	Inter-token gap tail: stalls
Out of memory as concurrency rises	KV cache, not math, caps the batch

Rapid-fire Q&A

Question	Crisp answer
Why is decode memory-bound?	One token per step reuses each weight byte once: ~1 op/byte vs ~300 needed.
What two things does every step read?	All the weights and the request's whole KV cache.
How does batching help?	The weight read is shared by every request in the batch.
What caps the batch?	KV memory: each request keeps its cache on the GPU.
How close to the floor am I?	Achieved bandwidth ÷ peak, e.g. 16.6 GB ÷ 12 ms ≈ 1.4 TB/s ≈ 41%.

TRAP

- Treating the floor as the expected speed: it's the ceiling.
- Trusting GPU utilization during decode: the math units can be idle.
- Averaging TPOT only: stalls hide in the gap tail.